Assignment 11
RAG & Vector Databases

Part C — Building GenAI Applications

21 questions

★	**Must-know** — asked very frequently in GenAI / ML interviews (11 questions). Prepare these first.
[CODE]	**Hands-on task** — write and run the code (3 tasks).

1. What is RAG? Explain its architecture (indexing pipeline + retrieval + generation) with a diagram. ★
2. RAG vs fine-tuning vs long-context prompting — when would you use each? ★
3. What are embeddings? Name OpenAI embedding models and open-source embedding models. Can you reduce OpenAI embedding dimensions? ★
4. How do you choose an embedding model (MTEB leaderboard, domain, language, dimensions, cost)?
5. What are chunking strategies (fixed, recursive, semantic, document-structure based)? How do you choose chunk size? ★
6. How are indexes created in vector databases? Explain flat, IVF, PQ and HNSW. ★
7. How does HNSW (Hierarchical Navigable Small Worlds) work? ★
8. Compare vector databases: Chroma, FAISS, Pinecone, Qdrant, Weaviate, Milvus, pgvector.
9. Store embeddings in ChromaDB, persist them to disk and load them back. [CODE]
10. Create retrievers that fetch: (a) top-5 records, (b) records with similarity score > 0.5, (c) results using MMR. [CODE]
11. What is MMR (Maximal Marginal Relevance) and why use it? ★
12. What is hybrid search (BM25 + vector)? Why does it help? ★
13. What is re-ranking? Explain cross-encoders and Cohere Rerank. ★
14. Explain MultiQueryRetriever, ContextualCompressionRetriever, ParentDocumentRetriever and SelfQueryRetriever.
15. What are query transformation techniques (HyDE, query rewriting, query decomposition)?
16. Build a complete retrieval chain (retriever + prompt + LLM) that cites its sources. [CODE]
17. How do you build RAG using an open-source LLM end-to-end?
18. How do you feed table data and image data to a RAG system?

19. What is Agentic RAG? What is GraphRAG? ★
20. Your RAG system gives wrong answers. How do you debug it step by step (retrieval vs generation)? ★
21. Explain some real-world RAG use cases.